Final Project Delivery Report and Oral Defence

EGH490-2 RESEARCH PROJECT 2

Assessment #	**1**
Unit Learning Outcomes Assessed	ULO1. Formulate a solution to an engineering problem using a process that displays rigorous research and investigation practice, and accounts for risk, ethics and sustainability considerations at a mastered level. ULO2. Practise professionally in the collaborative environment of an engineering project at a mastered level. ULO3. Report a project delivery that addresses an engineering problem using written communication and defend that delivery using oral communication at a mastered level.
Task Weight (%)	60% (Assessment 1A - 40% and Assessment 1B 20%)
Due Date	Week 13, 11:59pm Wednesday (Assessment 1A: Project Delivery Report Written submission). Oral presentation (Assessment 1B) will be in Week 13 and detail arrangement will be announced through Announcement.
Nature of Submission	Written Report, incorporating: • Title page, formatted according to the supplied template • Maximum 20 pages incorporating the main body of the report (see details below) • References • Appendices Oral presentation: 15 minutes allocation, incorporating 10 to 12 minutes of presentation and 3 to 5 minutes of answering questions.

Overview

Engineers report the outcomes of engineering projects to other members of the engineering team and to clients and stakeholders. In this assessment you will produce a written report on your delivery of your capstone engineering research project and an oral defence highlighting the key areas of delivery and approach in undertaking the project. You will be questioned at the conclusion of your oral presentation to justify and defend your engineering decisions and outcomes. You should refer to the Engineers Australia Stage 1 Competency Standard for Professional Engineer to provide context for this aspect of engineering practice.

What you will do

You will prepare a Written Report on the delivery and outcomes of your Capstone Engineering Research Project. Following submission of this written report, you will then give an Oral Presentation with a summary of the methodology and outcomes from your capstone engineering research project. Following your presentation, you will answer questions from a panel of academic supervisors and fellow students. There may be invited professional engineers from industry in attendance.

Your written report should incorporate sections relevant to your project to convey the work you have completed across the year in delivering on your capstone engineering research project. It must include a Title page in the format provided, main body, references and appendices.

Title page: use the supplied template for Title page

The main body of your report must include:

- *Abstract*: precise and concise overview of the capstone engineering research project report. An abstract must highlight all essential findings about the project, including data. It should be possible to read the abstract and gain all key findings without reading the full report.
- *Table of contents*: list all headings and subheadings used,

- *Introduction:* research question or problem statement, incorporating the importance and significance of the project, aim(s) and 3 to 5 objectives of the capstone engineering research project. Provide the aim(s) concisely in one to two sentences (consult with your supervisor if necessary) and the objectives that support the aims in bullet form.
- *Literature review:* synthesis of previous research related to the topic and methodology of your research project. Identifying the gaps which your project has addressed in one paragraph and as a subsection.
- *Methodology:* **describe** the methods used to address the research questions and why these are appropriate to your project, outline any stakeholder engagement, design, simulation, data collection or testing approaches used and interpretation, validation, and analysis techniques. Modelling work requires a detailed description of assumptions. Include any risk, ethics, and sustainability considerations which you accounted for in your research project.
- *Results, Analysis, and Discussion:* present the outcomes including any stakeholder feedback, engineering designs, simulation outcomes, data analysis, and performance data. Discuss these in relation to the gaps identified in the literature review and explain their significance. Briefly address the outcomes of your consideration of risk, ethics, and sustainability. Identify any limitations in your study.
- *Conclusions:* summarise the key findings and any recommendations for further work

References: list all sources that have been referred to within the research project report, including research publications, regulations or codes, or technical and conference reports.

Appendix 1 Research Project Timeline: include a summary of how the project was completed, including interim milestones and progress achieved across the two semesters

Other appendices: include materials which support your research project report but are too detailed to include within the main body.

Your oral presentation should present a summary of your capstone engineering research project approach and outcome and will be in person on campus. You will be allocated 15 minutes for your presentation, which includes 10 to 12 minutes for presentation and 3 to 5 minutes for questions. You need to carefully structure and organise your presentation and manage your time. Your audience will be a panel of academic supervisors and fellow students and may include professional engineers from industry. Consider ways in which you can engage and inform your audience. Your presentation should include:

- A brief description of the context for the project, aims and objectives (in one slide which must match the text in the report), background literature and the research gaps.
- A short overview of your research project approach and outcomes (methodology, results, conclusions). You may use text, diagrams, images, graphics, or visualisation to convey your message. Choose this with consideration of your audience.
- Your reflections on the significance of your research project outcomes, particularly with regards to other work you have identified via your literature review.
- Your reflections on any limitations, challenges or difficulties of the research project approach and outcomes, and recommendations for future work or lessons learned.

Your presentation should make effective use of presentation technology and give due consideration to the audience, including choosing fonts, images, colours, and other visuals that clearly convey your message.

What you will submit/present (what artefacts will be used to judge your performance?)

Your written report should meet conventions found within the Engineering Report Style Guide, formatted to print on A4 paper using a font size of 12 points for the main text, using a standard font (Arial, Calibri or Times New Roman). Margins are to be: top 20mm, bottom 20mm, left-hand side 20mm, and right-hand side 20mm, with headers and footers within these margins.

You should prepare your oral presentation in a format that can be uploaded to the QUT Standard Operating Environment for presentation in the room you have been allocated to. Your presentation slides should be submitted via the unit Canvas site prior to your allocated presentation time. You are strongly encouraged to practice your presentation and become familiar with the presentation technology before your scheduled presentation time.

Criteria for Assessment

There are four criteria that will be used to assess your submission:

1. **Engineering knowledge (weight: 20 marks) –** extent to which research project demonstrates knowledge of contemporary engineering research practice incorporating clarity of formulation of research aims and objectives, synthesis of relevant literature and identification of gaps, appropriateness of methodology, validity and reliability of analysis and conclusions.
2. **Engineering research application (weight: 20 marks) –** application of systematic engineering techniques, tools, and resources to an engineering research project to achieve outcomes; appropriateness of consideration of risk, ethics, sustainability, and stakeholder engagement.
3. **Professional communication: Written (weight: 10 marks) –** effectiveness of written communication in presenting engineering work, that is succinct, justified, and clear including appropriate structuring of report, and effective use of textual, diagrammatic, pictorial, and graphical media suited to the project context.
4. **Professional communication: Oral (weight: 10 marks) –** choice of presentation media, structure of presentation and quality of delivery with regard to clarity, timing, and audience engagement.

The attached marking rubric outlines the standards required to achieve a grade against each criterion.

Criterion 1 and Criterion 2 will draw on evidence from the written report, oral presentation, and your answers to questions at the end of the presentation.

Criterion 3 will draw on evidence from the written report.

Criterion 4 will draw on evidence from the oral presentation and answers to questions.

Note that in class lectures and meetings with your supervisor will provide you with details of how your mark will be determined.

Resources and Support Available

Lectures from EGH490-1

Week 5: Professional & Technical Communication: Writing

Week 11: Professional & Technical Communication: Presentations

Support

Library: consult with Faculty Librarians

Student Success: Book an appointment or attend a workshop at qut.to/academicsupport

Resources

Engineering Technical Report Style Guide

Library Guide: **https://libguides.library.qut.edu.au/engcap**

Check the unit Canvas site for other useful resources related to writing and presentation.

Academic Integrity Statement

You are reminded that maintaining academic integrity is an important professional capability in all assessment. See the QUT MOPP C/5.3 Academic Integrity and the Engineers Australia Code of Ethics and Guidelines on Professional Conduct. For this assessment academic integrity will be monitored via:

- Interactions with your supervisor where you should discuss the details of your project and the preparation of your report and presentation.
- Plagiarism detection during submission via Turnitin.
- Questions from a panel of markers and fellow students.

In finalising your mark, you may be asked to verify aspects of your submission through an interactive oral. Note that if the Unit Coordinator has reasonable concerns about your academic integrity, they may ask you to authenticate your learning in line with MOPP C/5.3 Section 5.3.7 (e) *Authentication of learning.*